

Q1. What is Exploratory Testing?

Exploratory testing is a process where test design ,test execution and logging works simultaneously.

in exploratory testing tester explore the application dynamically by using their experience to identify defects.

Exploratory testing is not random testing but it is adhoc testing with the purpose of finding bugs.

Exploratory testing is structured and functionalities are checked in an ad-hoc manner.

Q2. What is a traceability matrix?

Traceability matrix is a document that maps and tracks the relationship between different software developments such as requirements ,test cases and defects and it ensures complete test coverage and helps verify that all requirements are tested.

- Types of traceability matrix

- 1) Forward traceability: Mapping of requirement to test cases

- 2) Backward traceability: Mapping of test cases to requirement.

- 3) Bidirectional traceability: Combine forward and backward traceability.

- Benefits of Traceability Matrix

- Ensures that the client that the software is being developed as per requirement.

- To make sure that all the requirement should be included in test case.

- If any changes in requirement we can easily find out which test case needs to be updated.

Q3. What is Boundary value testing?

Boundary value testing is a process to identify the defects at the boundaries of input.it works on those points when input values change from valid to invalid errors.

Benefits of boundary value testing:-

- Reduce test case while maintaining effectiveness

- Helps to detect boundary related defects.

- For an exam.

- System Accept Values from 1 to 100.

- Valid boundaries are 1 to 100.

- Invalid boundaries are 0 to 101.

- Test cases are 0,1,99,100,101

Q4. What is Equivalence partitioning testing?

Equivalence partitioning testing is a black box technique that divides input into groups and selects one input to test them all and it reduces the test cases.

- Benefits of Equivalence partitioning testing:-

- Reduce test cases while maintaining good coverage.

- Saves time and efforts in large scale testing

- For an exam.

- System allows user to register their age is between 18 to 16.

- Valid partition: 18 to 60.

- Invalid partition: Less than 18 is invalid input.

- Invalid partition: Greater than 60 is invalid input.

Q4. What is Integration testing?

Integration testing is a process where the two individual units combine and are tested as a group. It's called integration testing.

- There are 2 levels of Integration Testing

- Component integration testing.

- Is performed after unit testing to verify individual components work together correctly when it's combined.

- Types of Component integration testing:

- functional testing and non-functional testing.
- System integration testing.
- It verifies the integration of the entire system with the external system.

- Integration Testing Methods

- Big Bang Integration Testing: all modules integrated once and tested together.
- Top Down Approach: Testing takes place from top to bottom and takes help of stubs for testing.
- Bottom Up Approach: Testing takes place from the bottom to the upwards and takes help of drivers for testing.
- Stub and Driver Approach: Act as a substitute for the missing models in the testing.
- Continuous Integration Approach: Used in a software development method where automated tests run frequently.

Q5. What determines the level of risk?

A factor that could result in future negative consequences and need to identify the risks associated with your project.

- Risks are of two types:

- Project Risk : it occurs when a senior team Member leaves the project abruptly.
- Product Risk : Example of product risks would software system not installing in test environment

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Q6. What is alpha testing?

Alpha testing is always performed by the developer at the software developer site but sometimes it's also performed by the independent testing team.

- Alpha testing comes under black box and white box testing.
- Alpha testing is conducted for software projects.
- Alpha testing performed in a virtual environment.

Q7. What is beta testing?

Beta testing is always performed when software products and projects are marketed. It is not performed by the independent testing team.

- Beta testing is conducted for software products.
- Beta testing comes under black box testing.
- Beta testing performed in a real time environment.

Q8. What is component testing?

Component testing is a process where an individual software component will be tested. It's also called unit testing.

- Component testing covers functional and nonfunctional testing.
- Component testing performed by using white box testing.
- Component testing finds defects at the early stage of the life cycle.

The purpose of the component testing is to validate that each unit of the software performs as designed.

Q9. What is functional system testing?

Functional system testing is a process where we have to evaluate the system compliance with its functional requirement. It ensures that the system functionality will meet the business requirements.

- There are two types of Test Approach
- Requirement based functional Testing : Testing against requirement and specification.
- Process Based Testing : Test procedures and cases derived from Expected user profiles, Business scenarios, Use cases.

Q10. What is Non-Functional Testing?

Non-functional testing is testing the attributes that do not relate to functionality of a system.

Non-functional testing is performed after functional testing.

Non-functional testing checks the usability, response time, performance, scalability, portability etc.

Non-functional testing is the testing of how the system works.

Q11. What is GUI Testing?

GUI testing is a process to evaluate the graphical user interface of the system and ensures that it functions according to design specifications.

Gui testing checks the elements like buttons, icons, text fields etc.

Approach of GUI Testing: manual based testing, record and replay, model based testing.

Q12. What is Adhoc testing?

Ad Hoc testing is an unscripted software testing approach where the tester tests the application without any test case or plans.

The aim is to find defects through random ways and it relies on the testers' experience.

-> Types of Adhoc Testing:

- Buddy testing: The tester and developer work together on the same module.
- Pair testing: Two testers work together on the same module.
- Monkey testing: Randomly test the application without any test case or plan.

Q13. What is load testing?

Load testing is a type of performance testing where it checks the system behavior under load. such as testing a web site under load and checking what point system degrades the response time or fails.

Identify bottlenecks, improve performance and ensure stability under high loads.

Q14. What is Stress Testing?

Stress testing is a type of performance testing where it checks the system under extreme conditions.

The aim is to stress test how the system behaves after the system fails.

The purpose of stress testing how the system recovers after the failure of a system.

Q15. What is white box testing and list the types of white box testing?

In white box testing the tester requires knowledge of the internal structure of a system.

- Types of white box testing
 - Statement coverage: it Covers Only true condition.
 - Decision/Branch Coverage: It covers true and false conditions.
 - Condition Coverage: This is closely related to decision coverage but has better sensitivity to the control flow.

Q16. What is black box testing? What are the different black box testing techniques?

In black box testing testers test the functional and nonfunctional without knowledge of the internal structure of the system. black box testing focuses on what a system does, rather than how it does it.

- Black box testing techniques :-
 - Equivalence partitioning
 - Boundary value analysis
 - Decision tables
 - State transition testing
 - Use case testing

Q17. Mention what are the categories of defects?

- Functional Defects.
- Non-Functional Defects.
- UI/UX Defects
- Integration Defects
- Regression Defects
- Security Defects
- Compatibility Defects
- Usability Defects
- Performance Defects

Q18. Mention what bigbang testing is?

Big bang is a type of integration testing Where all modules and components of a system are combined and tested all together in one go.

- Convenient for small systems.
- Useful when all modules are dependent on each other for functionality.
- Difficult to debug because all modules are tested together so hard to identify root cause of defects.

Q19. What is the purpose of exit criteria?

Purpose of exit criteria is to define when we stop testing.

- End of all testing products go live.
- End of phase of testing
- All highly prioritized bugs are fixed and closed.

Q20. When should "Regression Testing" be performed?

Regression testing is performed when new code is implemented in an application and checks it does not affect the existing functionality.

It ensures that old code still works once the new code changes are done.

Q21. What are 7 key principles? Explain in detail?

7 key principles help the testers efficiency,testing efforts,and enhance software quality.

- Testing Shows Presence of Defects: Testing can show that there are defects but cannot prove that there are no defects present.
- Exhaustive Testing is Impossible: there are multiple combinations of inputs and preconditions so testing everything is impossible. so we have to prioritize our testing efforts using a risk based approach.
- Early Testing: Testing Should start as early as possible in the software development life cycle.
- Defect Clustering: small no of modules Contains most of the defects and that time the defects does not spread evenly into the system because they are clustered.

- Pesticide Paradox: If the same test repeats again and again the test case no longer finds any new defects to find new defects testers should regularly update, modify or create new test cases.
- Testing is Context Dependent: Testing an application depends on its type so different kinds of applications are tested differently.
- Absence of Errors Fallacy: If the build of the system does not fulfill the requirements of the user it doesn't make it a good system.

Q22. Difference between QA v/s QC v/s Tester?

Quality Assurance	Quality Control	Testing
1) Quality assurance is establishing and improving process to ensure quality.	Quality control inspecting and Testing the final product.	Testing identify the defect/error/bug In the system.
2) Focus on process and procedure rather than actual testing.	Focus on actual testing by executing software.	Focus on actual testing.
3) Process oriented activities.	Product oriented activities.	Product Oriented
4) It is a subset of STLC.	It is a subset of QA	It is a subset of QC

Q24. Difference between Smoke and Sanity?

Smoke Testing	Sanity Testing
1) Smoke testing checks the functionality of a system.	Sanity testing Checks the new functionality of a system.
2) Smoke testing performed by the developer and tester.	Sanity testing Performed by the tester.

3) Smoke testing is usually documented and scripted.

Sanity testing is usually undocumented and unscripted.

4) Smoke testing is a subset of regression testing.

Sanity testing is a subset of acceptance testing.

5) Smoke testing checks the entire system from component of end to end.

Sanity testing checks the particular a system.

Q25. Difference between verification and Validation?

Verification	Validation
1) The process of ensuring the software is being developed according to the requirements.	The process of ensuring the final product meets the user requirements.
2) Are we building the product right?	Are we building the right product?
3) Plans, test cases, codes, requirements.	The actual product.
4) Reviews, inspection.	Testing.

Q26. Explain types of Performance testing?

- Load testing: Load testing is a type of performance testing where it checks the system behavior under load.
- Stress testing: Load testing is a type of performance testing where it checks the system behavior under extreme conditions.
- Endurance testing: Endurance testing is a type of performance testing that evaluates how a system performs under a sustained load over an extended period.
- Spike testing: Spike testing is a type of performance testing that evaluates how a system handles the sudden and extreme increases in load.

- Volume testing: volume testing evaluates the systems performance by handling a large amount of data.
- Scalability testing: scalability testing is a type of performance testing that evaluates a system's ability to handle increased workload by scaling up without compromising performance.

Q27. What is Error, Defect, Bug and failure?

- Error: when a mistake is found by the developer it's called an error.
- Defect: when an error is found by the tester it's called a defect.
- Bug: when a defect is accepted by the developer it's called a bug.
- Failure: when a build does not meet the requirements it's called a failure.

Q28. Difference between Priority and Severity?

Severity	Priority
1) Make the order in which defects should be fixed as per requirement of business.	Indicates the impact of defect on the application's functionality.
2) Severity is absolute and Customer-Focused.	Priority is Relative and Business-Focused.
3) Types: low,medium,high,critical.	Types: major,minor,moderate,cosmetic,critical

Q29. What is the Bug Life Cycle?

Bug life cycle is a process that tracks the progress of a defect from its discovery to closure and ensures that the bugs are resolved in an organized way.

- Stages of Bug life cycle:
 - New
 - Assigned
 - Open -
 - Fixed
 - Pending retest
 - Retes

- Verified
- Closed

Q30. Explain the difference between Functional testing and NonFunctional testing?

Functional testing

- 1) Functional testing ensures system functionality will meet the business requirements.
- 2) Functional testing is executed first.
- 3) Functional testing describes how product works.
- 4) Types of Functional testing are:

Unit Testing
Smoke Testing
Sanity Testing
Integration Testing
White box testing
Black Box testing
Acceptance testing
Regression Testing

NonFunctional testing

- Non functional testing checks the reliability and other non-functional aspects of the system.
- Non functional testing should be performed after functional testing
- Nonfunctional testing describes how good the product does.

Types of Nonfunctional testing:

Performance Testing
Load Testing
Volume Testing
Stress Testing
Security Testing
Installation Testing
Compatibility Testing

Q31. What is the difference between the STLC (Software Testing Life Cycle) and SDLC (Software Development Life Cycle)?

STLC	SDLC
1) STLC is used for testing a developed software. software.	SDLC is used for developing
2) STLC is focuses on verifying and validation of software.	SDLC is focused on overall creation that the software works correctly.
3) Start when requirements are defined. initiated.	Starts as soon as the project is
4) Ends when testing are completed and quality maintained.	Ends when software is deployed and criteria are met.

Q32. Explain what the Test Plan is? What is the information that should be covered?

A test plan is a detailed document that describes how the testing will be carried out for a particular project.

- Information that should be covered:
 - Preparation of test plan.
 - Strategy document for various types of testing.
 - Test tool selection.
 - Test effort estimation.

Q33. What is priority?

Indicates the impact of defects on the application's functionality and Priority is Relative and Business-Focused.

- Types of Priority:

- Major
- Minor
- Moderate
- Cosmetic
- Critical

Q34. What is severity?

Make the order in which defects should be fixed as per requirement of business and Severity is absolute and Customer-Focused.

- Types of severity:

- Low
- Medium
- High
- Critical

Q35. Bug categories are...

- Functional bugs
- Performance bugs
- Usability bugs
- Security bugs
- UI bugs
- Regression bugs

Q36. When to use Usability Testing?

- Early : detect major usability issues early when changes are easy.
- Before release: Check validate usability.
- After release: catch real world issues.

Q37. What is the procedure for GUI Testing?

Manual based testing: In manual based testing with graphical interface should be checked manually by tester and checks the requirement met to the business requirement.

- Record And Replay: GUI testing can be done using automation tools.
- Model Based Testing: A model is a graphical description of a system's behavior. it helps us to understand and predict the system behavior

Q38. Write a Scenario of Lift.

- What type of cable is used in the lift for moving?
- what is the capacity of the lift in weight & person.
- What type of material is used to build lift.
- Lift door is automated or manual.
- Lifts are built for how many floors.
- Lifts have telephonic connections or not.
- Lift has emergency button or not.
- Lift has a alarm or not.
- Lift has a stop button or not.
- What's the speed of lift?
- Lift CCTV or not.
- Lift has a smoke detector or not.
- Lift has emergency exit or not.
- If lift automated doors are not working can we operate manually.
- Floor button matches the actual floors.
- Lift has an open close button.
- Floor no. is displayed or not.
- Lift has lights and fans.
- Lift is use for commercial purpose or civil.

- Lift is moved to particular floor while floor button pressed.
- Lift is usable for blind people.
- Check if the sensor is working properly or not.
- Lift has an overloading alert or not.
- Lift is moving or not while the doors are open.
- Lift has audible functionality or not.

Q39. Write a Scenario of Pen.

- What is the type of pen ball, gel or ink?
- What is the color of ink in a pen?
- What is the type of the body of the pen.
- What is the mechanism of a pen?
- If mechanism of pen is spring check it works properly or not.
- If pen is ballpoint pen check the ball point is working properly.
- Check whether pen has refill or use and throw.
- If the pen has an ink check, the ink doesn't leak.
- Check the cap of the pen fits perfectly.
- Check the grip of the pen and verify if you are able to hold a pen.

Q40 Write a Scenario of Pen Stand.

- What is the material type of pen stand?
- Check whether pen stand is use for single pen or multiple pens.
- A pen stand is able to hold the weight of pens.
- Check the height of the pen stand.
- Check pen stands made for which purpose.
- Check what the shape of the pen stand is.

Q41. Write a Scenario of Door.

- What is the material type of door?
- If the door type is wood, check if it's waterproof or not.
- If the door type is wood, check if it's insect proof or not.

- If the door type is wood, check What type of wood is used to make the door.
- Check the thickness of the door.
- If the door type is metal check it has rust proof properties.
- Verify that the dimensions of the door are as per the specification.
- Check if the door opens in an inward direction or outwards direction.
- Check if the door is a sliding or rotating door.
- Check if the door is closing perfectly or not.
- Verify if the door has stopped or not.
- Verify that we can close the door from inward as well as outward direction.
- Check the types of locks in the door.
- The Check Door mechanism works perfectly.

Q42. Write a Scenario of an ATM.

- Check display functionality.
- Check if all buttons work properly.
- Check speakers of atm works properly.
- Check the cash dispense mechanism works properly.
- Check receipt print mechanism works properly.
- Verify the card reader functionality works properly.
- Verify the camera of the atm works properly.
- Verify the cash deposit mechanism works properly.
- Verify the button works properly as per functionality.
- Verify the microphone functionality works properly.
- Verify the height and length of the atm is as per specification.

Q43. Write a scenario of Microwave Oven.

- Check the functionality of the microwave oven.
- Verify the buttons of the microwave works properly.
- Verify that the microwave heats food properly.
- If the microwave has rotational temperature control functionality, check if it works properly.
- Verify timer of microwave is working properly.
- Verify the rotational functionality of the microwave works properly.

- Check if the power cable of the microwave works properly.
- Check the microwave door to open and close properly.
- Check the timer display works properly.
- Check the capacity of the microwave.
- Check what type of container is appropriate for a microwave.
- Check the maximum and minimum heating capacity of the microwave.

Q44. Write a scenario of a Coffee vending Machine.

- Check the functionality of the machine.
- Verify the outer body and inner parts of the machine work properly.
- Verify the quantity of the machine.
- Verify the heating functionality works properly.
- Verify the cable of machine works properly
- Verify the vending machine is digital or manual.
- Verify the machine is leak proof.
- Verify the serving size of the machine.
- Check the display functionality works properly.
- Verify the machine button works properly.
- Check if the ingredients work properly.

Q45. Write a scenario of a chair.

- Check the material type of chair.
- Check the weight capacity of the chair.
- Check the use of the chair.
- Check the height, weight and length of the chair as per requirement.
- Check the stability of the chair.
- Check if the cushion is provided or not as per requirement.
- If the chair type is wooden, check if it's waterproof or not.
- If the chair type is metal, check if it's rust proof or not.
- If the chair type is plastic, check the durability of the plastic.
- If the chair has a folding mechanism, check if it works properly.

Q46. Write a Scenario of Wrist Watch.

- Check the material type of watch.
- Check the watch perfectly fits on hand as per requirement.
- If the wrist watch is digital, the check display works properly.
- Verify the time set functionality works properly
- If wrist watch is analog the no. is align perfectly.
- Check if the inner mechanism of the watch works properly.
- If the watch has metal material, check if the lock mechanism works properly.
- If the watch has date and day functionality, check if it works properly.
- If the watch is analog, check the details of the watch.
- Check if the watch is waterproof or not as per requirement.
- Check whether the wrist watch is visible or not.
- Verify the time format of the watch.
- Check whether the watch has a battery slot or not and whether it's replaceable.

Q47. Write a Scenario of whatsapp Group (generate group).

- Check the purpose of making a group.
- Check how many members are there as per requirement.
- Check how many members can send a message or only the admin can send a message.
- How many admins are there in the group?
- Check the photo photoworks properly.
- Check the emoji option works properly.
- What is the duration of disappearing messages?
- Check group permissions button works properly.
- Check we can add members after creating a group.
- Check call and video call functionality works properly.
- Check the edit name of group functionality works properly.
- Check remove members functionality works properly or not.
- Check report and exit group functionality works properly.

Q48. Write a Scenario of Whatsapp payment

- Verify if the payment button is clickable or not.
- Verify that you ask for the amount written or not.
- Verify that for the amount written there is a typing pad or not.
- Verify that the typing pad is clickable or not.
- Verify that after entering the amount the next button is working or not.
- Verify that the upper side of your bank name is showing or not.
- Verify that the upper side whom to send money to is showing or not.
- Verify that after clicking the next button it shows both information or not.
- Verify if the payment method is change or not.
- Verify that the view account balance option is work or not.
- Verify that the send payment button is working or not.
- Verify that the UPI PIN option is work or not.
- Verify that continue button is work or not

Q49. Write a scenario of only Whatsapp chat messages.

- Verify that opening any chat works or not.
- Verify that closing any chat works or not.
- Verify that send messages works or not.
- Verify that images & videos are sent works or not.
- Verify that images & videos are sent in different types of work or not.
- Verify that sending emojis works or not.
- Verify that sending stickers works or not.
- Verify that sending gifs works or not.
- Verify that create avatar works or not.
- Verify that send avatar works or not.
- Verify if the change is working or not.
- Verify that text editing works or not.
- Verify if the shared Gboard works or not.
- Verify that floating works or not.
- Verify that the resize works or not.
- Verify if the next language works or not.

- Verify that undoing works or not.
- Verify that the send location works or not.
- Verify that the audio sent works or not.
- Verify that AI images work or not.
- Verify that poll is works or not
- Verify that the contact sent works or not.
- Verify that the audio message is recorded and sent or not.
- Verify that sended message confirmation is shown or not.
- Verify that disappearing messages work or not.
- Verify that after a long press on the space button, change language works or not.
- Verify that the message forward works or not.
- Verify that sended message sent date & time shows or not.
- Verify if the pin message works or not.
- Verify that sended message delete or delete for everyone is works or not.
- Verify that create and add to this list is works or not.
- Verify that Clear chat works or not.
- Verify that block contact works or not.
- Verify that the react message works or not.
- Verify that sended message can give a star i.e. works or not.
- Verify that any message's reply function is works or not